NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATICS

DEPARTMENT OF INFORMATICS

QUALIFICATION: BACHELOR OF COMPUTER SCIENCE, BACHELOR OF INFORMATICS, BACHELOR OF COMPUTER SCIENCE IN CYBER SECURITY, BACHELOR OF COMPUTER SCIENCE IN SOFTWARE ENGINEERING, BACHELOR OF COMPUTER SCIENCE IN SYSTEMS ADMINISTRATION AND NETWORKS		
QUALIFICATION CODE: 07BACS, 07BAIF, 07BCCS		**LEVEL:** 5
COURSE NAME: MATHEMATICS FOR COMPUTING AND INFORMATICS 1B		**COURSE CODE**: MCI521S
DATE: 15 SEPTEMBER 2026		**MARKS:** 100

ASSIGNMENT 1

INSTRUCTIONS

1. Answer **ALL** the questions and number your answers clearly and correctly.
2. Marks will not be rewarded for answers obtained without showing the necessary steps leading to them (the answers).
3. Write clearly and neatly in dark blue or black ink. Any work written in pencil will not be marked.
4. Indicate your mode of study (Part-time or Full-time.
5. This is an individual assignment, to be submitted as a hard copy to your respective lecturer.

THIS PAPER CONSISTS OF 3 PAGES (including this front page)

Question 1 [14 marks]

1.1 Given a system of linear equations:

$$\begin{aligned} px + qy + rz &= u \\ sx + ty + wz &= v \\ mx + ny + pz &= w \end{aligned}$$

Using the Gauss Elimination method, show that:

$$z = \frac{(qm - pn)(sv - pu) - (qs - pt)(mv - pw)}{(qm - pn)(rs - pw) - (qs - pt)(rm - p^2)}$$

$$y = \frac{1}{qs - pt}[(sv - pu) - z(rs - pw)]$$

$$x = \frac{1}{p}(u - yq - zr)$$

Question 2 [72 marks]

2.1 Find the value of p for which the vectors $(3, 1, -2)$ and $(-2, p, 4)$ are orthogonal (perpendicular). [3]

2.2 If $(x, 2, 1) + 3(2, y, -1) - 2(-1, 4, z) = (9, 1, 4)$, what are the values of x, y,and z?

[3]

2.3 Given that $u = (3, -2, 1)$, $v = (2, 1, -4)$, and $w = (1, 0, -2)$. Find $\|3u - 2v + 4w\|$ [4]

2.4 Find the values of the scalars a, b, c such that

$(5, 8, 3) = a(1, 1, 0) + b(0, 2, 1) + c(2, 0, 1)$ [5]

2.5 Solve the following system of equations using Cramer's rule [10]

$$\begin{aligned} 2x + y - z &= 3 \\ x - 2y + 3z &= 9 \\ 3x + 2y + z &= 16 \end{aligned}$$

2.6 Solve the system of linear equations for x, y and z using Gaussian Elimination: [10]

$$x + 2y + 3z = 14$$

$$2x + 5y + 2z = 18$$
$$3x + y + 4z = 15$$

2.7 Find the matrix D such that $ADB = C$ and hence evaluate $f(D^{-1})$ given that [12]

$$A = \begin{pmatrix} 1 & 4 \\ -2 & 3 \\ 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 8 & 6 & -6 \\ 6 & -1 & 1 \\ 4 & 4 & -4 \end{pmatrix} \; and \; f(x) = -x^3 + 3x^2 + 2.$$

2.8 Find the inverse of the matrix [10]

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{pmatrix}$$

2.9 Find the determinant of the adjoint of the matrix [15]

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 4 & 5 \end{pmatrix}$$

Question 3 [14 marks]

An electronics manufacturer produces two types of components: Circuit Boards and Microcontrollers. Each Circuit Board requires 3 hours of soldering time and 2 hours of quality inspection. Each Microcontroller requires 4 hours of soldering time and 1 hour of quality inspection. The factory has a maximum of 120 hours available for soldering and 50 hours for inspection per week. Warehouse space limits the total combined output to no more than 35 units per week. The profit per Circuit Board is N$20, and the profit per Microcontroller is N$30.

3.1 Formulate a linear programming model. [5]

3.2 Determine how many units of each component should be produced to maximize total weekly profit. [6]

3.3 Calculate the maximum total weekly profit. [3]

END OF ASSIGNMENT